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Q. 1 Assume that you are assigned responsibility of developing an *Online Date Sheet Generation System* (ODSGS) for a University. *ODSGS* should run both on PCs and Mobile Devices. *ODSGS* will have fields such as Course Code, Course Title, Date of Exam, Time of Exam etc. The user of *ODSGS* will input the list of course codes and course titles and duration of examination. Sunday will be holiday. Starting date of exams will also be input by the user. There may be other information that is needed by *ODSGS* to generate the date sheet. So, you can list them and make provision for them in database structure. Make assumptions wherever necessary. Based on the inputs received, *ODSGS* should generate the Date Sheet. For developing *ODSGS* as specified above,

(a) Which SDLC paradigm will be selected. You may also suggest a SDLC paradigm that is proposed by you and non-existent as on date. Justify your answer.

Ans. SDLC is a process followed for a software project, within a software organization. It consists of a detailed plan describing how to develop, maintain, replace and alter or enhance specific software. The life cycle defines a methodology for improving the quality of software and the overall development process.

The Spiral model seems as an ideal choice here. The spiral model is similar to the incremental development for a system, with more emphasis placed on risk analysis. The spiral model has four phases: Planning, Design, Construct and Evaluation. A software project repeatedly passes through these phases in iterations (called Spirals in this model).

Spiral Model

No other model seems a reasonable alternative to accept as a different answer. This model combines the features of the prototyping & the waterfall model. As Online Date Sheet Generation System (ODSGS) is a large project, therefore spiral model is intended for large, complex, expensive & complicated projects. The steps in the spiral model can be generalized as follows:
1. The new system requirements are defined in as much detail as possible. This usually involves interviewing a number of users representing all the external or internal users and other aspects of the existing system.

2. A preliminary design is created for the new system.

3. A first prototype of the new system is constructed from the preliminary design. This is usually a scale down system, and represents an approximation of the characteristics of the final product.

4. A second prototype is evolved by a fourfold procedure:

(1) Evaluating the first prototype in terms of its strengths, weaknesses, and risks;
(2) Defining the requirements of the second prototype;
(3) Planning and designing the second prototype;
(4) Constructing and testing the second prototype.
Strength:
1. Estimates (i.e. budget, schedule, etc.) become more realistic as work progresses, because important issues are discovered earlier.
2. It is more able to cope with the (nearly inevitable) changes that software development generally entails.
3. Software engineers (who can get restless with protracted design processes) can get their hands in and start working on a project earlier.
Weakness:
1. Highly customized limiting re-usability
2. Applied differently for each application
3. Risk of not meeting budget or schedule
4. Risk of not meeting budget or schedule
Advantages:
1. Estimates (i.e. budget, schedule, etc.) become more realistic as work progresses, because important issues are discovered earlier.
2. It is more able to cope with the (nearly inevitable) changes that software development generally entails.
3. Software engineers (who can get restless with protracted design processes) can get their hands in and start working on a project earlier.
Disadvantages :
1. Highly customized limiting re-usability
2. Applied differently for each application
3. Risk of not meeting budget or schedule
4. Risk of not meeting budget or schedule

Software development life cycle (SDLC) (also referred to as the application development life-cycle) is a process for planning, creating, testing, and deploying a software system. The system development life cycle framework provides a sequence of activities for system designers and developers to follow. It consists of a set of steps or phases in which each phase of the SDLC uses the results of the previous one.

The SDLC adheres to important phases that are essential for developers–such as planning, analysis, design, and implementation. Like anything that is manufactured on an assembly line, an SDLC aims to produce high-quality systems that meet or exceed customer expectations, based on customer requirements, by delivering systems that move through each clearly defined phase, within scheduled time frames and cost estimates.

SDLC done right can allow the highest level of management control and documentation. Developers understand what they should build and why. All parties agree on the goal upfront and see a clear plan for arriving at that goal. Everyone understands the costs and resources required.

(b) List the functional and non-functional requirements.

Ans.

Functional requirements

In software engineering, a functional requirement defines a function of a software system or its component. A function is described as a set of inputs, the behavior, and outputs (see also software). Functional requirements may be calculations, technical details, data manipulation and processing and other specific functionality that define what a system is supposed to accomplish.

User Interfaces:

1. Home screen
2. menu selection screen
3. Online Date Sheet Generation System (ODSGS) Form Online
4. Online Date Sheet Generation System (ODSGS) Instruction information
5. Online Date Sheet Generation System (ODSGS) Form Offline
6. Online Date Sheet Generation System (ODSGS) form Status
7. Download

Configuration:

Minimum 2GB Hard

Disk P-III processor or equivalent Ram 512 MB Windows with Apache preloaded.

Client Configuration A terminal with Internet Explorer and Printer. Software Interfaces Operating system – WindowsXP, OracleNetwork -- LAN

Non-functional requirements :-

Performance Requirements System can withstand even though many number of users requested the desired service. As we are keeping office level server of the automated payroll system and access is given to the only registered users of office who requires the services of viewing, Updating etc. It can withstand the load. Security Requirements Sensitive data is protected from unwanted access by users appropriate technology and implementing strict user access criteria. Software Quality Attributes Menu-driven programs with user friendly interface with simply hyper links. It is very easy to use. Backup mechanisms are considered for maintainability of software as well as database. As it is object oriented reusability exists. As project is based on MVC architecture, testability exists. Safety & Reliability Requirements By incorporating a robust and proven SQL into the system, reliable performance and integrity of data is ensured. There must be a power backup for server system.

(c) Estimate the cost.

Ans. Software Cost Estimation is widely considered to be a weak link in software project management. It requires a significant amount of effort to perform it correctly. Errors in Software Cost Estimation can be attributed to a variety of factors. Various studies in the last decade indicated that 3 out of 4 Software projects are not finished on time or within budget or both who is responsible for Software Cost Estimation? The group of people responsible for creating a software cost estimate can vary with

each organization. However the following is possible in most scenarios – People who are directly involved with the implementation are involved in the estimate. - Project Manager is responsible
for producing realistic cost estimates. - Project Managers may perform this task on their own or consult with programmers responsible. - Various studies indicate that if the programmers responsible for development are involved in the estimation it was more accurate. The programmers have more motivation to meet the targets if they were involved in the estimation process. For Factors contributing to inaccurate estimation Scope Creeps, imprecise and drifting requirements · New software projects pose new challenges, which may be very different from the past projects. Many teams fail to document metrics and lessons learned from past projects. Many a times the estimates are forced to match the available time and resources by aggressive leaders. Unrealistic estimates may be created by various „political under currents" Impact of Under-estimating: Under-Estimating a project can be very damaging It leads to improper Project Planning - It can also result in under-staffing and may result in an over worked and burnt out team Above all the quality of deliverables may be directly affected due insufficient testing and QA Missed Deadlines cause loss of Credibility and goodwill.

(d) Estimate the effort.

Ans. Estimating The process of forecasting or approximating the time and cost of completing project deliverables. The task of balancing the expectations of stakeholders and the need for control while the project is implemented the two primary elements in test estimation are time and resources. Your estimation needs to take both into account. There are many questions you need to answer in order to do test estimation. The more accurate and thorough your answers to these questions the better your test estimation. What modules or functionalities will be tested and how many testers are available to test them? Of course, as functionalities increase and/or number of testers decrease the more time it will take to thoroughly test the application. What is the complexity of each of these modules or functionalities? As the complexity increases the more time and effort will be required to For. understand the application create test plans create test cases execute test cases regress test cases and retest defects. How many test iterations (test runs) will be required to complete the test project? This is also related to complexity. As an application becomes more complex it will typically require more test iterations to reach the company's exit criteria (the number of open defects by severity and priority that a company can live with). How much time will be required by developers to produce fixes for new builds between test runs? Complexity is also a factor here. As an application becomes more complex there are often more dependencies between modules and functionalities. This often requires coordination between developers. Consequently, this takes more time. This is important because your estimation must also include the amount of time testers are waiting for the next build between test runs. What is the average number of defects that you anticipate will be found during each test run? You may have already guessed that complexity is a factor here too. The more complex an application the greater number of defects will reach the test team when the application is released to them. For. In addition, the more complex the application the more likely that severe and high priority defects will be found in later stages of the test process.

(e) Develop SRS using IEEE format.

Ans. Preface this document contains the Software Requirements Specification (SRS) of an Online Date Sheet Generation System (ODSGS). The main aim of this project is to add functionality to the existing SUMS system in order to provide an online facility for managing and registering student accounts and generate Date Sheet. This document has been prepared in accordance with the IEEE Std 830-1998, IEEE Recommended Practice for Software Requirements Specifications [IEEE 1998].

1. Introduction This Software Requirement Specification is written accordance with the IEEE Std 830-1998 model.

1.1. Purpose For this SRS Document contains the complete software requirements for the Online Date Sheet Generation System (ODSGS) and describes the design decisions, architectural design and the detailed design needed to implement the system. It provides the visibility in the design and provides information needed for software support.

1.2. Scope Online Date Sheet Generation System (ODSGS) used to replace old paper work system. SAS is to build upon the existing web-based project marking system PUMS in order to implement the project marking process and allocating supervisor/ideas to students. This increase in efficiency of project marking, audit trails of marking process, give feedback to student, finally, publication and email student result. It provides a mechanism to edit the online marking form which makes the system is flexible.

1.3. Definitions Acronyms and abbreviations Online Date Sheet Generation System (ODSGS) For. PUMS Project Units Management System SRS Software Requirements Specification SUMS Student and Units Management System J2EE Java 2 Platform Enterprise Edition JSP Java Server Page UP Link UOP Student Portal Facility OS Operating System

1.4. References Brigg s Briggs, J. (2005). SUMS documentation. Retrieved 3 rd December 2005, For .2005 from http://www.tech.port.ac.uk/staffweb/briggsj/jimapp/S UMS/ IEEE 1998 IEEE Std 830-1998, IEEE Recommended Practice for Software Requirements Specifications. ISBN 0-7381-0332- 2.

1.5. Overview This document has been prepared in accordance with the IEEE Std 830-1998, IEEE Recommended Practice for Software Requirements Specifications [IEEE 830-1998 (1998)]. It provides the information of Product perspective, Product functions, User characteristics, Constraints, Assumptions and dependencies and specific requirement.

2. Overall description This section of the SRS describes all general factors of the product and its requirements.

2.1. Product perspective for.

2.1.1. System interfaces The SUMS is the new updated version of PUMS - the web-based project unit management system. It is intended to implement all PUMS's features for the administration of student projects. The SUMS is using J2EE platform and Struts Model 2. All components follow Model-View- Controller pattern. SUMS import JimApp packages that can either connecting to an Oracle database or MySQL database through the Database Utility components. The possible extension is to inter-connection to UP Link System which provide student with many functions, including the ability to check assessment results. Students can connect both systems to retrieve information on their academic progress.

2.1.2. User interfaces All pages of the system are following a consistent theme and clear structure. The occurrence of errors should be minimized through the use of checkboxes, radio buttons and scroll down in order to reduce the amount of text input from user. JavaScript implement in HTML in order to provide a Data Check before For submission. HTML Tables to display information to give a clear structure that easy to understand by user. Error message should be located beside the error input which clearly highlight and tell user how to solve it. If system error, it should provide the contact methods. The page should display the project process in different colour to clearly reflect the various states that student done. Each level of user will have its own interface and privilege to manage and modify the project information such as supervisor able to monitor/manage his student progress and make comment on it,
student can change his detail, view the progress, submit project idea. The System should provide a feedback form for all users to give comments or asking questions. It should provide a FAQ to minimize the workload of system administrator.

2.1.3. Hardware interfaces
a. Server Side The web application will be hosted on one of the department's For. Linux servers and connecting to one of the school Oracle Database server. The web server is listening on the web standard port, port 80.

b. Client Side The system is a web-based application; clients are requiring using a modern web browser such as Mozilla Firebox 1.5, Internet Explorer 6 and Enable Cookies. The computer must have an Internet connection in order to be able to access the system.

2.1.4. Software interfaces

a. Server Side The UOP already has the required software to host a Java web application. An Apache Web server will accept all requests from the client and forward SUMS specific requests to Tomcat 5.5 Servlet Container with J2EE 5.0 and Strut 1.2.8 hosting SUMS. A For. development database will be hosted locally (using MySQL); the production database is hosted centrally (using Oracle).

b. Client Side An OS is capable of running a modern web browser which supports HTML version 3.2 or higher.

2.1.5. Communications interfaces The HTTP protocol will be used to facilitate communications between the client and server.

2.1.6. Memory The UOP already hosts a number of Java web applications, it is not anticipated that OPMS will require any additional memory storage. g) Operations Procedures are already in place as part of the UOP's IT policies for data security and Backup. For .h) Site adaptation requirements. There should no site adaptation requirement since the Web Application Server was setup and running Java web application.

2.2. System functions This section outlines all the main feature of SAS.

2.2.1. Student role The Student can register a SUMS account and start the progress of project. On the register form, student should enter all their detail such as HEMIS numbers, Email and contact number. The system will generate activation code and send email to student and confirm the registration. After, the system allow student to change information and provide the function forget password for student to retrieve back the password.

2.2.2. Administration role The system administrator must be able to:

1. deactivate and reactivate student account login For.
2. force the sending of a new password to a student via email
3. change any of a student's details

2.2.3. Audit Trailing Each user will have an associated record of history. This will provide information on various events such as Previous Development - a number of components have been developed by the client, Jim Briggs, and previous developer, Steven J Powell. New components need to compatible to the exist system.

2.5 Assumptions and dependencies Although basic password authentication and role based security mechanisms will be used to protect OPMS from unauthorized access; functionality such as email notifications are assumed to be sufficiently protected under the existing security policies applied by the University network team. Redundant Database is setup as the role of backup Database Server when primary database is failure. For. The correct functioning of OPMS will partly be dependent on the correctness of the data stored and managed as part of the PUMS system. Also, the application will be hosted by the UOP as one of many applications; the event of the server failing due to an error with one of these applications might result in SAS becoming temporarily unavailable.

3. Specific requirements

3.1. Functional requirements

3.1.1. User class - Student This section is for UOP School of Computing Student

1. Student registration form. Student can be register on the system and fill in all detail and forward to choose project/supervisor.
2. Change Detail. Student can change detail if information is For. Incorrect such as telephone number.
3. Change Password. Student can change his login password at any time for security reason.
4. Forget password. Student can request his password if he/she forgotten the password.

3.1.2. User class - Academic Staff All staff can view the details of any student.
3.1.3 User class - Unit Cohort co-ordinator Certain staff may be designated as Unit or Cohort Coordinators and can change the details of any student doing their unit or project cohort. Change Student Detail Unit Cohort co-ordinator can change student detail for incorrect information. View Student Detail Unit Cohort co-ordinator can view student information and monitor their progress. For. List Student Unit Cohort co-ordinator can list all students in different period of different group. Change Password Unit Cohort co-ordinator can reset the student's password if required.
3.1.4 User class - System Administrator List Student System Administrator can list all students in
different period of different group to check any error. Change Password System Administrator can reset the student's password if required.
3.1.5 User class - Administration Staff List Student Administration Staff can list all students in different period of For. different group.

3.2 Design constraints The system need to design base on the existed code and database using J2SE 5.0, J2EE 1.4 and Struts 1.2.x.

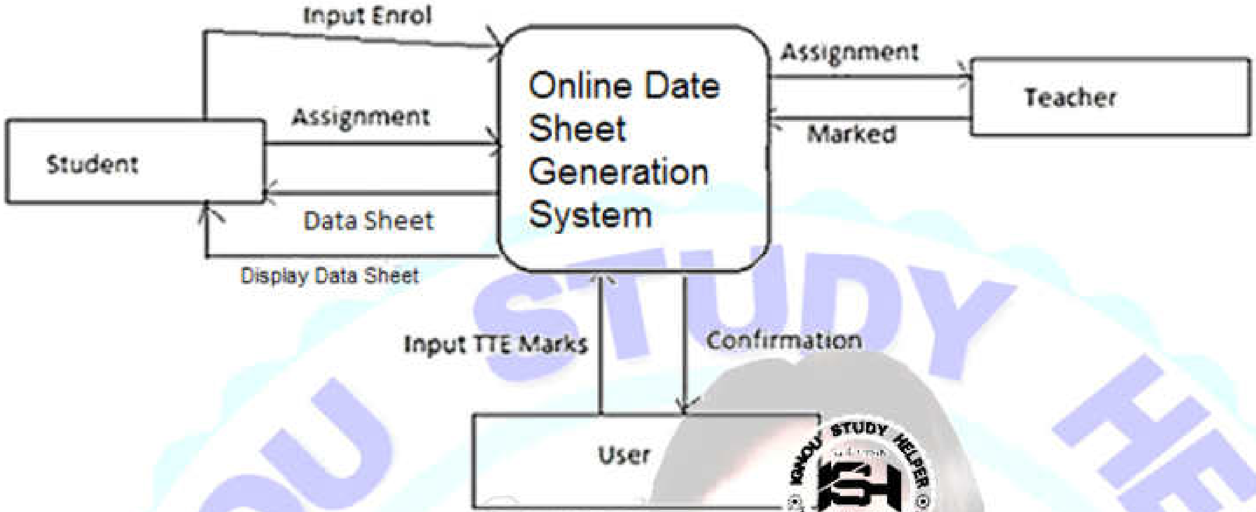
3.3 Software system attributes.
3.3.1 Security The system needs to log client's information of registration such as IP address and time for security purpose. Password should encrypted and store in the database.

3.3.2 Maintainability The system developing using Struts, all action is detailed in struts-config.xml and web.xml that easy to modify and make update.

3.3.3 Portability The web application is coding in J2EE and Struts, therefore, it For. should be transferable between different OS and Java container.

3.4 Other requirements For further extending, is able to send notification by SMS.

DFD


Input Enrol
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Online Date Sheet Generation System
Assignment
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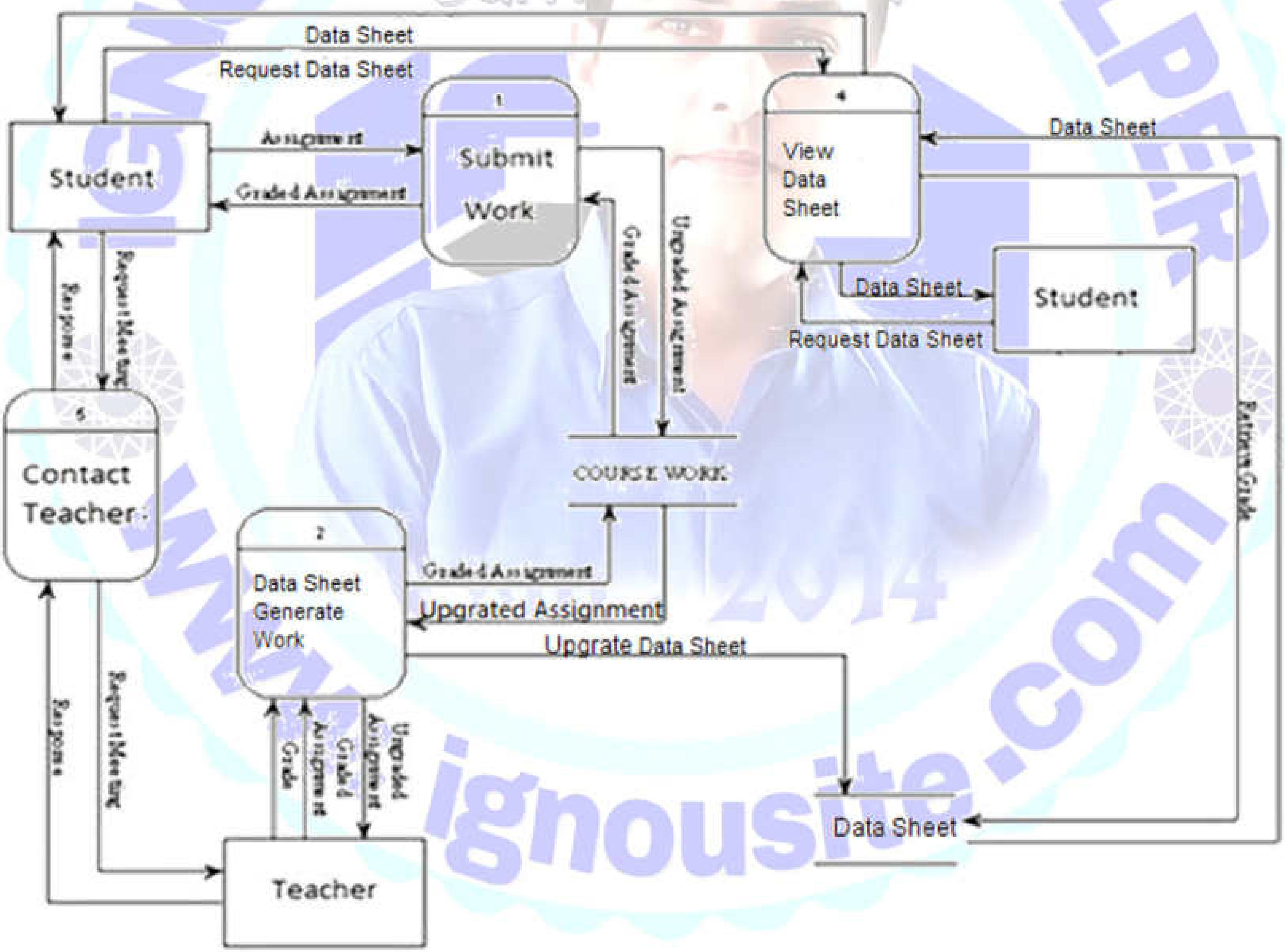
Data Sheet
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View Data Sheet
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Request Meeting
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5
Contact Teacher
COURSE WORK
2
Data Sheet Generate Work
Graded Assignment
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Response
Request Meeting
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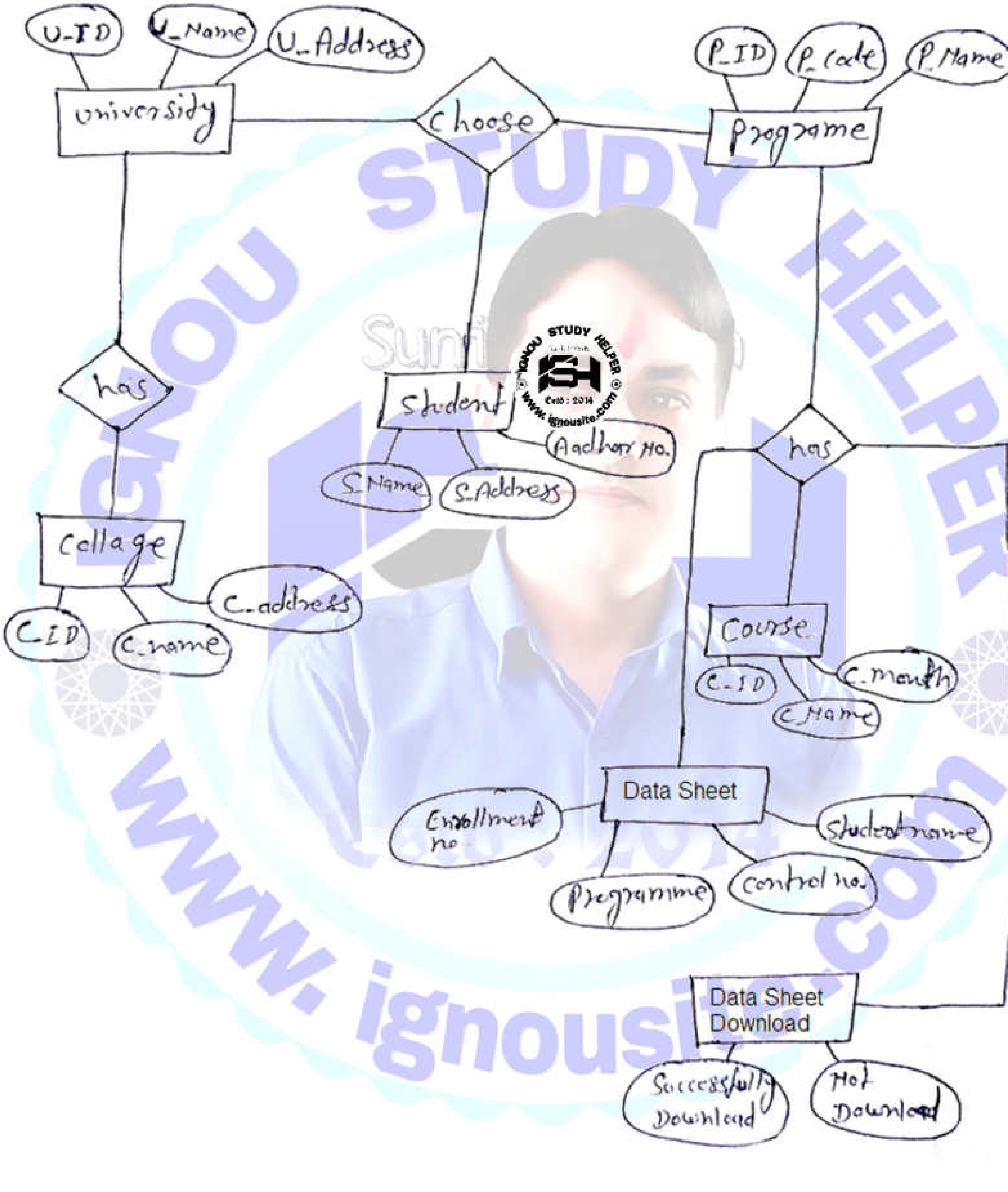
Upgraded Assignment
Data Sheet
Teacher

ER- Diagram


U-ID
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Data Sheet Download
Successfully Download
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(f) List queries for whom Reports can be generated.

Ans. This module handles all the configuration and administration of the system. Online Date Sheet Generation System (ODSGS) admin module has following features.

The admin department does a lot of work in the complete admission gimmick. They are actual technology providers in the entire admissions. There are multiple technical functions, the admin has to perform. These functions help in the smooth functioning of the integrated admissions.

The enlistment of the functions of an Admin is:

- Grievance management
- Telephone and email support
- Report downloading
- Information Access
- Report Compliance
- Category-wise Generate Data Sheet
- Gender ratio
- Institute Analysis

A. Quota Table Management:

Admin can manage quota table according to rules and regulations of the government entity. Quota table includes details of total available seat for individual college along with stream, male female and other reservation details. Admission allocation is done according to this quota table.

B. Student Data Management:

All the applications data is available on single click which can be sorted/ verified/ exported to CSV / XML formats as per the requirement Student data is used to generate admit letters, analysis purpose.

C. Merit List Generation

According to Govt Rules and Regulations and Weightages Merit list is generated for valid applicants based on criteria defined for the Online Date Sheet Generation System (ODSGS) process.

D. Change Requests:

Students can send their misc change requests through their login & admin can accept / reject the changes & open the submitted form for correction & re-submission.

E. Student Grievance Management:

Any student query/ correction in Name/Category can be communicated by the student which can be managed from admin panel.

The admin has to address and respond to the complaints and queries of the candidates. If the candidate is stuck at any point in the admissions, say he is not able to upload his photograph or something, he can contact the support in such case so that he can get the desired help.

Grievance can be resolved in the centralized Online Date Sheet Generation System (ODSGS) to correct information and resolve errors associated with Online Date Sheet Generation System (ODSGS) data, merit list etc.

Quick resolution of student grievance can fast track overall admission process and it can be completed within stipulated deadline.

F. Report Downloading

Every college is supposed to generate their own report of Data Sheet which includes the statistics of number of Data Sheet made, the enrollment no., Student name on Data Sheet, programme name and so on.

This report is supposed to be shared with the admin so that even the admin has the data for his functioning. This data management is extremely essential as it becomes easier for the admin to keep track of the smooth working of the Data Sheet on his end.

G: Information Access

At any certain point, if there is any problem with the accessibility of any part of the software at any end of any entity, that's where the admit comes to the rescue.

He/ She can provide access to that particular entity through his login. Another aspect of Information access also means that he has access to every mentioned information, may it be of the candidate, the college, or any other entity.

H: Report Compliance

The admin is supposed to generate reports of the entire procedure. The aim to generate this report to conform the data provided and available. It also turns out to be a helping hand when it comes to improving the system for future. Another objective behind this report enlisting is also that the colleges and universities can work on their part to make enhancements if any needed.

I: Category-wise Data Sheet

The term 'Category' doesn't necessarily mean the programme category, but also the course wary. The admin allots courses that are specified by the student/candidate. This too has different aspects like the student's eligibility and course availability at collaborated colleges.

The programme wise Data Sheet means admitting students according to their reservation quotas in the education system. There are certain number of seats allotted to every college for each and every stratum.

J: Gender Ratio

The gender ratio analysis comes under the few above listed functions like Category-wise programme, Report Compliance, etc. The admin has to scrutinize the male female ratio of Data Sheet allotment. This is for some certain institutions who give special privileged Data Sheet to either the makes or the females.

K: Institute Analysis

The term is similar as the report generation. This is essentially needed when it comes to granting Online Date Sheet Generation System (ODSGS). The keywords in the report analysis consist of the basic Data Sheet procedure like, Data Sheet made, the enrollment no., Student name on Data Sheet, programme name per course etc.

This report creation is an attempt to endorse the colleges or institutions to make elevations down the line.

(g) List specific requirements which enables ODSGS to run on both PCs and Mobile Devices.

Ans. Some specific requirements which enables ODSGS to run on both PCs and Mobile Devices under this, well-structured progress of phases is visible that each organization focuses on very seriously. This allows them to create high-quality software after testing the software rigorously and then making it ready for manufacturing.

There are essentially six phases to this methodology, and they are as follows:

- Analysis of requirement
- Planning the development
- Design of the software, like the architectural design
- Developing the software
- Conducting tests
- Deployment of software

Project Initiation: - After you have decided on which project to go forward with and signed the agreement, the next phase is initiating it. This is when the entire software development life cycle starts, and you have to figure out the project goal.

Thereafter, the team has to state the success criteria associated with the project and its goal. In case you are under a T&M contract with the company, this phase is still important to go through.
The next point of concern in this phase of an SDLC Example is to make a Project Charter. In that, you have to prepare and define which factors would work with the project in question. After that, you have to figure out who the key stakeholders are for the process.
There are many points of the process where they would give their input, such as:-

- User Experience
- Hardware Specifications
- User Interface
- Technical Limitations
- Software architecture, and
- The environment of the development

While you can get feedback on such points from the customers and clients, that is not enough. Having expertise from within your team and outside would help you create the perfect Software Development Projects Examples. After making out the project motto and aim, you can jump to developing the concept.

Concept Development:- Indeed, developing software is not a one-way road; there are different manners of creating web applications. One of the most important and useful ones in this regard is to develop and design the concept first.
For the application that you wish to create, you have to make a conceptual wireframe and design. If the detailing is not set, that is still fine; as long as you have a basic framework to refer to in later stages.
Of course, there is a great reason why you should focus on this stage more carefully, especially regarding the stakeholders. Having a concept design ready would give them a more substantial project to focus on, and they would engage more.
Not all the stakeholders would understand the more complicated development language. Thus, such simplified imagery would help them realize the end-product better.
Aside from the design part of the concept development stage, the technicalities are also important in an SDLC Example.
This is because not all the projects can work with the already available solutions and needs more specific requirements. In such a situation, your team has to create a whole new concept that would bring the correct outcome you desire.
Here, you can take the help of the best experts in this field to exchange ideas with, and figure out unique solutions. In case you cannot find the correct solution after brainstorming, you have to continue with the present approaches.
And if you need more time and resources to work the solution better, your Software Development Projects Example may get canceled. Overall, this stage is often termed as the gateway stage.

Planning:- After clearing the concept, the next obvious step is to create a plan. If the project is a large one, you have to create a completely full-fledged plan from the first step. Thus, it is better to start off with smaller projects if you are a beginner.
It is more simple and easy to handle and plan. However, it is important to mention here that most teams focus on creating a customized approach to each project management program.
Thus, you can go forward with using the already present approaches, like Kanban and Scrum variations. Or, you can make a completely customized plan, instead. As for the main points that are covered during the SDLC Example planning, they are as follows.

- Figuring out the scope of the project
- Settling milestones
- Estimating the costs and time required
- Identifying the expertise and resources that you would need also comes under this particular phase.

Therefore, by the end of this SDLC phase, you may have committed to delivering on time the product in its complete form. Or, you would have created the plan for project management. In the case of the former, you would have to figure out how to work the project as per the organization's customs.

Requirements :- The next of the Product Development Life Cycle Stages out what you would need to complete it. To do that, you can adopt the following methods.

- Hold interviews
- Brainstorming with the team
- Create and distribute surveys and questionnaires
- Mind mapping
- Depend on focus groups
- Notice wireframes
- Analyze the relevant documents
- Listen to user stories

You do not need to follow any particular step only to gather requirements. As long as the software engineers understand what you need as requirements, you can continue making the SDLC with Example.
You can tweak the requirements and make it more suitable for future projects too.
Following this, focus on implementing the high-level practical analysis of the project with the requirements.

Designing :- At the design stage of the software development life cycle, you would notice two specific aspects.

- UI and UX Design
- Technical Architecture

In this portion of working through SDLC Real Life Example, you have to probe the previously gathered requirements. Following that, you need to figure out the UI or User Interface design that would make the service or application functional.
Plus, you need to create an architecture that can hold all the requirements in a way that would work for them.
The designs that you would develop are in the following manners.

- Workflow diagrams
- Wireframes
- Mockups
- Listed Frameworks, Technologies, and Libraries
- Documented description of the Architecture

Following this, you can get started with the coding process that would work toward the proper development of the product.

Development :- After planning, gathering, and designing, the next step of the Software Development Life Cycle Template is executing it.
The programmers would start with the coding process on a day-to-day basis. Selecting the framework of your choice, you have to continue with the development procedure at this point.
You have to prepare a working prototype or model of the application and show it to the stakeholders. That would increase their engagement and create a better quality product.
They would suggest changes or more specifications that you need to implement in the SDLC Example. At this point in the life cycle, you have to consider their points and repeat and increment the project continuously.

However, while this stage may demand more time, you still have to deliver the end-project on a scheduled date. The deadline and budget that the client specified cannot get changed, and you have to keep that in mind.

Integration :- All the applications nowadays integrate completely with other services, and you have to focus on giving your application the same functionality.
This is especially important in companies, as their employee information is generally synced to another database.
Thus, you need to focus on integrating your software with these databases, so that you can access their storage space.
In terms of what services to integrate the SDLC Example with, one name is the third-party providers. These include Cloud applications like Dropbox, among others.
Also, you have to make your software soluble with the market you are selling at and the business processes of the client company.
There is a lot of collaboration that is involved at this stage, which makes it very time-consuming. Thus, you should check for the requirements and possible risks for integration from the beginning.
While it is part of the development process, it involves a lot of complicated elements. Owing to this, treating it as a phase in an SDLC Example is most suitable.

Testing :- In the final stretches of the development process, it is important to handle the final testing. Not only is testing important as a separate stage but also it is necessary at the other stages too.
If you handle this throughout the beginning stages, that would help you detect and correct any issues beforehand. If you test it well and find no defects in the program, that ensures Quality Assurance.
Even if there are some minor defects in the program, people can still use it. However, you should know the latest Software Testing Trends to give a perfect result. But you have to check and see if they are of working condition and provide the list of problems during certification.

Deployment :-

During the deployment period of the Product Development Life Cycle Stages, you have to deliver it to a hardware.
Whether it is a mobile device, PC, server, or MAC, you have to use the software on these devices. Overall, the main purpose at this stage is to make the software you developed accessible to end-users.
For deployment on desktops or phones, you have to prepare an installer for your software to work with them. Plus, you have to provide this installer in mobile application stores like Google Play or App Store.
On the other hand, if you are deploying it to a server, you have to upload the application directly into it. Then, the next step for a programmer is to connect this application to other servers and services.

Maintenance :- The maintenance of the application occurs throughout the Development Life Cycle Stages, but sometimes, you would notice a shorter period of it.
This usually appears after deployment, when some development team members keep working on the application still. During this period, they look for any critical issues that previously left their notice, and they fix it.
Later, after some months of verification and user testing, the support work is passed to a different support team.

Closure, Hand-off, and Support :-In the last phase of the SDLC Example, you still need to put in more effort. You have to gather everything about the application you created and its specifications and pass it to another support team.
You need to identify many other requirements, though these you should prepare beforehand. Plus, you have to provide extra documents, what more capabilities are needed later, etc. However, like the identification of the requirements, you have to understand and plan the hand-off process beforehand.